Research on cock states detection of cock oven inspection robot based on YOLO-v3

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***Abstract*—The state switching of the coke oven gas pipeline exchange cock plays a vital role in the safe production of coking. This paper designs and manufactures a rail-type mobile robot applied to coke oven basement inspection tasks to troubleshoot coke oven heat exchange equipment. Three detection methods are proposed for the opening and closing state of the exchange cock. Each detection method is analyzed according to the actual production process and requirements. The cock images are collected by the robot and made into a dataset. YOLO-v3 is used to train models of three detection methods on the same experimental platform and compare them. Finally, the optimal detection model is selected and deployed on the inspection robot. The detection accuracy rate reaches 99.1%, which effectively reduces the accident rate and maintains the stability of the coking system.**

***Index Terms*—inspection robot, coke oven, cock recognition, object detection, YOLO-v3**

I. Introduction

The coking industry is an important process industry that converts coal into metallurgical fuels and chemical raw materials. Coke oven equipment is the foundation of the coking industry. How to maintain equipment safety and troubleshoot timely is an important research direction. The gas pipelines of the modern horizontal coke oven are distributed in the basement at the bottom of the coke oven, and the air heat exchange system is neatly arranged in the basement and flue corridor [1].

Coke ovens are used for high-temperature production all year round, with frequent heat exchanges. However, coke oven basement equipment is mostly manually inspected, and workers frequently enter the first-level explosion-proof area containing dangerous gases. They need to bear high operational safety risks and pose health risks. At the same time, it is difficult to concentrate on the patterned and high-intensity work, which reduces the accuracy of inspections.

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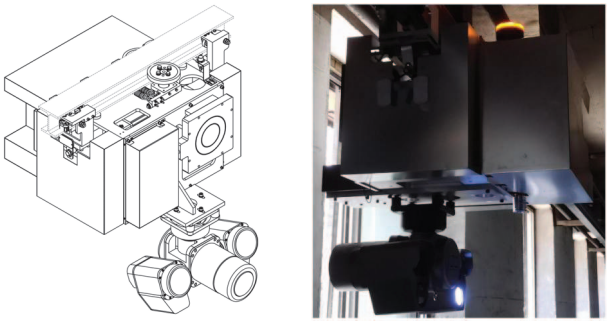
With the vigorous development of robots and computer technology, intelligent inspection robots have become the focus of domestic and foreign enterprises. The use of robots to replace manual inspections can not only reduce labor costs, but also provide accurate detection accuracy and improve work efficiency. Timely investigation of equipment failures plays a very important role in improving the safety and maintenance capabilities of coke oven site management and ensuring the safe and effective operation of the coking plant [2].



(a) (b)

Fig. 1. (a) Robot design drawings. (b) Robot objects.

Inspection robots have been gradually used in coke oven production in recent years. Gehui Xie from Taiyuan University of Technology designed the mechanical structure of a coke oven temperature measurement robot [3]. Guoqiang Pan proposed a method to lift the furnace lid in response to the height limit requirements of the temperature measurement robot [4]. A roller-type lifting mechanism is used, and solutions are proposed for key issues such as heat dissipation and control of the robot. Shi Yingdi who from ACRE, designed a rail-type inspection robot system for coke oven basements [1]. Lu Jianwen from Baogang Group proposed a new method for testing coke oven ventilation equipment [2]. A recognition algorithm based on three-dimensional point cloud, binocular vision and OpenCV vision library is proposed.

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The above research applied to the coke oven basement does not involve the detection plan of the gas pipeline exchange cock. This paper is oriented by industrial needs, based on the coke oven gas exchange cock inspection task, comparing various detection strategies, and finally chooses a coke oven gas pipeline exchange cock opening and closing state detection method based on YOLO-v3 [5]. After verification, it is deployed in the orbital inspection robot shown in Fig.1 has been put into practical application.

II. Robot Inspection process

The mechanical structure and pipelines in the basement of the coke oven are scattered and arranged. They are exposed to high temperature and cinder ash all the year round, which makes the surface of the mechanical structure eroded, and the color of the equipment is relatively similar to the background. The complex background and unstable light have caused a certain degree of interference to the detection of the cock.

During coke oven production, the equipment status is exchanged every 20 minutes. At the beginning of the exchange, the hydraulic system first drives the pull rod to close the opening gas pipe cock on the second floor underground. Then it switches the lifting and lowering of the mound rod and the opening and closing of the damper on the first floor underground to adjust the amount of air entering the combustion chamber. Finally, open the gas cock on the second floor underground that was closed during the last heat exchange process, and the equipment status exchange is completed. It can be seen that the exchange process needs to strictly control the working state of the exchange cock. Once the cock fails, it will lead to serious safety accidents such as gas backflow, gas leakage, pipeline combustion and even explosion.

Since the inspection task needs to be completed in a heat exchange process, the orbital inspection robot needs to walk about 150 meters in 20 minutes. Considering the wear of the wheel train and the power loss of the robot, the inspection robot in this paper adopts mobile photography.

The robot inspection process is shown in Fig.2. Firstly, the host computer issues inspection instructions, and the robot that receives the task walks on the track after completing the self-inspection. The robot records the furnace number corresponding to the current point through the RFID radio frequency card and triggers the camera command, and the binocular pan-tilt detects each group of exchange units. The visible light camera of the binocular gimbal uses Hikvision's "iDS-2ZMN3209N(B)" movement to identify the opening and closing state of the cock. Infrared thermal imaging uses the "D883NT" movement of Dali Technology to measure the temperature of the gas pipeline and prevent the gas inside the pipeline from burning. After temperature measurement, image acquisition, image processing, image storage and other steps, the obtained equipment comprehensive information is compared with the standard equipment information provided by the coke oven PLC. If a fault occurs, the error information and the fault location will be fed back to the host computer for alarm, and a supplementary inspection task will be generated.

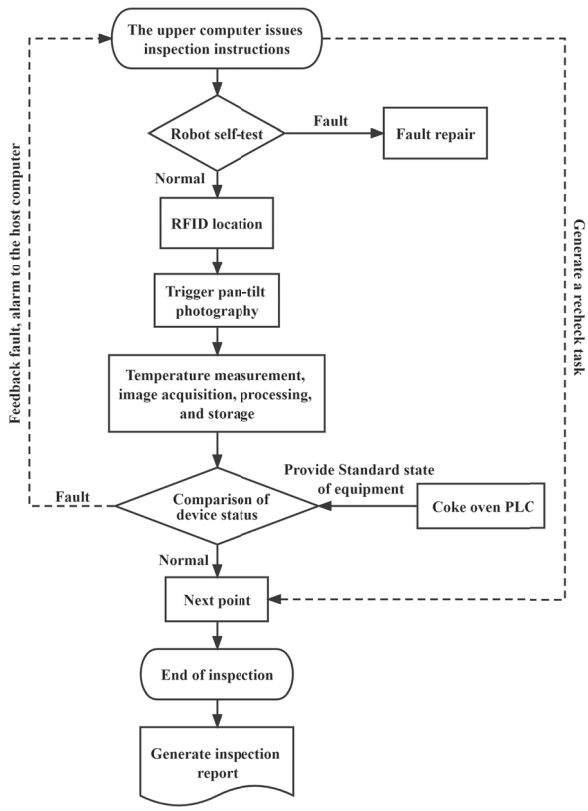

Fig. 2. Robot Inspection process.

The error point will be checked repeatedly before the inspection task ends. If the equipment is normal, the robot advances to the next point for detection. When the inspection is over, an inspection report is generated for workers to check.

III. Cock detection strategy

The coke oven gas pipeline is arranged on one side of the second floor underground of the coke oven basement. A group of machine-side exchange units consists of two coke oven gas pipelines. It can be seen from Fig.3 that the gas pipeline are given two colors of red and green in the coking process, and the "T-sign" mark is engraved on the exchange cock core of each pipeline to distinguish the open and closed state of the cock. This inspection platform takes the manual inspection of "T-sign" as the starting point, uses a binocular pan-tilt to detect the "T", and uses this as the basic standard for subsequent experiments.

A. Detect "T-sign"

The content of the manual inspection is to detect the "T-sign" of the cock cores of the two pipes respectively, and the detection objects are divided into 4 categories: red pipe open, red pipe close, green pipe open, green pipe close (hereinafter referred to as red open, red close, green open, green close). However, compared with human eye recognition, it is difficult for a computer to directly identify the "T-sign".

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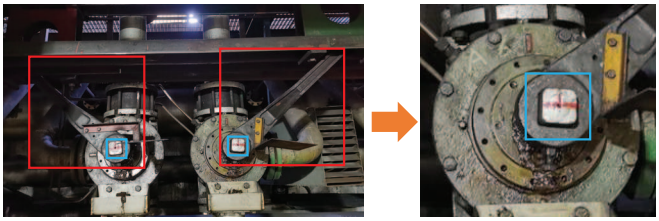

Fig. 3. Cock wrench and ”T-sign”

First of all, the target is small and the surface of the cock core is severely eroded. Eroded part of the ”T-sign” display is incomplete, and the quality of the image taken during the movement is difficult to meet the recognition requirements. Secondly, the cock core is often coated with lubricating oil, and the semi-solid lubricating oil mixed with coal ash and cinders can easily cover the mark. At the same time, the lighting conditions in the basement are unstable. The inspection robot must turn on the supplementary light when performing tasks, and the cock core may produce strong reflections. Furthermore, the ”T-sign” features are very similar, and only by identifying its direction, it is easy to cause misjudgment in a complex environment. Therefore, it is necessary to transform the feature of ”T-sign” into a dominant feature that is more conducive to recognition. Furthermore, the ”T-sign” features are very similar, and the direction is difficult to use as an explicit feature to distinguish targets in complex environments. However, in order to comply with industry inspection rules, the accuracy of this detection method is set as the basic standard.

B. Separate detection cock wrenches

Every 20 minutes, the pull rod pushes and pulls the cock wrench to switch the cock open and close state. It is easy to know that there is a unique correspondence between ”T-sign” and the direction of the wrench from Table I. In another word, there is a unique relationship between the direction of the cock wrench and the states of the cock. Therefore, using the wrench direction as the detection object during the inspection process will obtain more intuitive features and reduce the interference of surface wear and lubricating oil caused by direct detection of ”T-sign”.

TABLE I
COCK STATE-WRENCH DIRECTION RELATIONSHIP

	11 o’clock direction	2 o’clock direction
Red pipe cock	close	open
Green pipeline cock	open	close

As shown in Fig.3, the pixel ratio of the wrench feature in the image taken by the robot is much higher than that of the ”T-sign” cock core. Small target detection in complex environments requires more challenges, so increasing the pixel ratio of the target is a good idea to improve detection accuracy. The detection content is consistent with the detection of ”T-sign” and is divided into four categories: red open, red close, green open, and green close. The opening and closing status of each pipe is relatively independent, maintaining the amount of information for detecting the ”T-sign”.

C. Combination detection cock wrenches

Although detection scheme 2 alleviates some environmental interference and increases the target pixel ratio, it still does not solve the problem of direction detection of similar targets. Direction is not an obvious feature that distinguishes two objects in target detection. After field observation and combined with the coking process, it is found that there are only three stable states in the production process of the coke oven gas pipeline exchange cock, just like Fig.4. They are red open and green close, red close and green close, red close and green open.

(a) (b) (c)

Fig. 4. Three stable states

The detection combination feature eliminates the interference of similar features of some cocks, and reduces the difficulty of identifying the direction of similar targets. However, since the minimum detection unit is two cocks. Once a failure occurs, the report will lack the failure details of the red or green pipe, the amount of detection information is reduced.

D. Comparison of Three Detection Strategies

A comprehensive comparison of the three inspection strategies is shown in Table II. This article will compare three detection methods on the same dataset and select the optimal strategy to apply to the coke oven gas exchange cock inspection task.

IV. DETECTION ALGORITHM BASED ON DEEP LEARNING

At present, object detection algorithms applied in industrial environments are mainly divided into two categories: traditional algorithms and deep learning.

A. Traditional algorithms encounter challenges

In traditional machine learning algorithms, image recognition needs to extract the features of the image to describe the image. The main purpose of feature extraction is to reduce the dimension of the original image, map the source image to a low-dimensional feature space, and obtain low-dimensional sample features that can best reflect the essence of the image and distinguish it [6]. Commonly used traditional image recognition algorithms include Viola Jones detector [7] [8], HOG (Histogram of Oriented Gradient) [9], DPM (Deformable Part Model) [10] and so on.

However, in the complex environment of the coke oven basement, the traditional algorithm has encountered great challenges.

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TABLE II
COMPARISON OF COCK DETECTION STRATEGIES

	Detection class	Direction detection quantity	Strengths and weaknesses
"T-sign" detection	green open, green close red open, red close(4)	4	(1) "T-sign" is difficult to identify; (2) Lack of the features of red and green pipelines.
Wrenches separate detection	green open, green close red open, red close(4)	4	(1) High misdetection rate of direction detection; (2) The state of the cock is relatively intuitive.
Wrenches combination detection	red open and green close red close and green close red close and green open(3)	2	(1) The difficulty of direction detection is greatly reduced; (2) lack of bug report details.

- Firstly, traditional image recognition methods have high requirements for image contrast, blurring, inconspicuous grayscale, and uneven grayscale will increase the recognition error. The temperature of the coke oven equipment is high, oil and cinders are attached to the surface, and the surface of the wrench and "T-sign" is severely worn, and the color is relatively similar to the background.
- Secondly, the ambient light is unstable, which has a certain impact on the detection accuracy of the moving inspection robot.
- Thirdly, the traditional algorithm needs to manually extract features, and the recognition effect on targets with similar features is not good. Due to the strong symmetry of the coke oven basement equipment structure, similar detection objects only have different directions, which increases the difficulty of traditional algorithm recognition.

B. Deep learning

Since Alex Net's proposal in 2012, neural networks have become popular in the field of computer vision due to their excellent feature extraction capabilities [11]. As an end-to-end feature extraction method, deep learning uses neural network structure and loss function instead of tedious manual feature extraction rules, better uses massive data to learn features, and more clearly depicts the rich internal information of data. When the number of network layers is deepened, the abstraction of input features is also continuously improved, thus strengthening the ability of network classification and prediction.

At present, the deep learning algorithm has not been able to identify the specific tilt angle of the target, but it can achieve a rough direction recognition effect by converting the direction detection problem into a classification problem [12]. As a classic one-stage target detection algorithm, the YOLO series is widely used in industrial production. The algorithm divides the input image into grids, and automatically generates Anchor boxes on each grid to detect the targets falling in the grid, cancels the generation step of the prediction box in the two-stage detection algorithm, and greatly improves the detection speed. YOLOv3 has been tested by a large number of production practices, and has been widely recognized by the industry for its high-speed, stable, and accurate detection methods.

In order to detect targets of different sizes in the input image, as shown in Fig.5, the YOLO-v3 algorithm with darknet53 as the backbone network divides the output into three scales of 13x13, 26x26, and 52x52 by controlling the convolution step size and the connect operation [3]. This detection mechanism enables small target detection, such as cock "T-sign" detection.

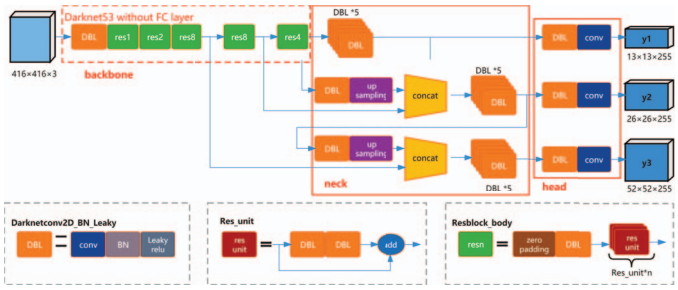

Fig. 5. Cock wrench and "T-sign"

V. EXPERIMENT AND DISCUSSION

Table III summarizes the model training platform and model deployment platform parameters. The data set images were collected by a rail-type inspection robot. According to the actual needs of the project, the robot moved forward at speeds of 0.1m/s, 0.13m/s and 0.15m/s under different lighting conditions. Images were taken while moving when passing inspection points. A total of 443 images of coke oven gas pipe cocks were collected, with the image pixel size being 1920×1080. The data set contains cock images under different speed tasks and different lighting conditions. The model will have certain robustness to interference conditions such as lighting and shooting blur.

TABLE III
EXPERIMENTAL PLATFORM

	CPU	GPU	System
Model training	Inter i9-10920X	NVIDIA-3080	Window 10
Model deployment	Inter i7-12700	NVIDIA-3060	Ubuntu20 x86_64

This article selects mAP (mean average precision), Loss, and FPS (Frames Per Second) as experimental evaluation indicators. Among them, mAP is the most noteworthy item in the target detection task, which represents the average accuracy

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of the model. In a high-risk environment such as a coke oven basement, the stability and detection accuracy of the detection system are closely related to production safety and worker life safety. Therefore, mAP was used as the main detection index in this experiment. The Loss value represents the difference between the model's estimated value and the actual value, and is an important basis for updating model parameters. A stable Loss indicates that the model has been regionally stable based on the training of the current data. FPS represents the number of images that the current model can process per second. Since the robot's inspection cycle is 20 minutes, there are certain requirements for the detection speed of each inspection point. Therefore, it is necessary to ensure the detection accuracy while selecting a faster model.

TABLE IV
MODEL EVALUATION

	IoU=0.5@mAP*	Loss	FPS
"T-sign" detection	89.63%	4.017	12.5506
Wrenches separate detection	97.41%	5.174	12.0100
Wrenches combination detection	**99.10%**	**2.243**	**13.0925**

* *Mean Average Precision.*

Set the same hyper-parameters to train the YOLO-v3 model. Table IV and Fig.6 show the performance changes and final evaluation results of the model training process. The combined detection wrench obtained the highest mAP, the lowest loss, and the highest FPS. The accuracy of "T-sign" detection is only 89.63%. External environmental problems such as reflection of the cock core and erosion of the logo affect the detection accuracy. In addition, this result proves that small target detection has limitations in complex environments. Using the rotation direction as a feature of target detection, the results are not ideal. The detection objects of manual inspection are not necessarily suitable for computer detection.

After making the "T-sign" feature explicit, the detection accuracy is significantly improved. The accuracy of detecting wrenches is 7.78% higher than that of detecting "T-sign", but the accuracy of 97.41% is not enough to meet the detection standards for high-risk environments in coke ovens. Since the similarity of the three types of targets in Figure 4 is much lower than that in Figure 3, the accuracy of the combined detection reaches 99.1%, proving that this strategy effectively solves the problem of difficulty in distinguishing red and green pipes due to the direction recognition of similar features in the above two solutions. At the same time, the combined detection wrench strategy has the fewest categories, performs better in terms of detection speed, and can meet the factory inspection speed requirements.

The wrench combination detection strategy detects large objects in complex environments, avoids the existence of multiple similar objects in the input image, and has certain robustness under unstable lighting conditions. The safety of coking production is the most important, and the robot should provide higher detection accuracy than manual inspection.

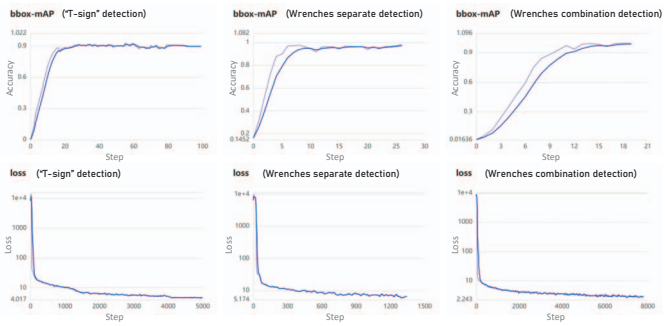

Fig. 6. Model performance

Therefore, this paper chooses the wrench combination detection strategy training model, and finally deploys it in the inspection robot and puts it into practical application.

VI. CONCLUSION

Based on the actual inspection needs of the coke oven basement, this paper designs and develops an intelligent inspection robot. Three detection methods were proposed for the coke oven gas pipeline exchange cock. After experimental demonstration and comparison, the optimal model was selected to be deployed on the robot and put into practical application.

The coke oven basement environment is complex and inspection tasks are heavy. The core idea of detection is how to eliminate environmental interference and amplify the characteristics of the detection target. In this paper, the identification feature of the cock core is extended to the state of the combined detection wrench, which greatly reduces the degree of on-site modification, uses the existing conditions of the equipment to seek the optimal detection result, and is proved to be effective in troubleshooting the cock, which is practical for maintaining the safe operation of the coke oven. It provides some experience for the subsequent development of coke oven inspection robots.

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